Hydroxide and oxides of beryllium are amphoteric in nature	Basic in nature.
It is not readily attacked by acids because of the presence of an oxide film	Readily attacked by acids
Beryllium carbide evolves methane with water.	evolve acetylene with water.
Salts of Be are extensively hydrolysed	Hydrolysed

Diagonal Relationship:

As observed in alkali metals, beryllium (the first member of group 2) shows a diagonal relationship with aluminium. In this case, the size of these ions ($r_{Be^{2+}}$ = 0.45 Å and $r_{Al^{3+}}$ = 0.54 Å) is not as close. However, their charge per unit area is closer (Be^{2+} = 2.36 and Al^{3+} = 2.50). They also have same electronegativity values (Be = 1.5; Al = 1.5).

Table 5.12 Similarities between Beryllium and Aluminium

S.No.	Properties
1	Beryllium chloride forms a dimeric structure like aluminium chloride with chloride bridges. Beryllium chloride also forms polymeric chain structure in addition to dimer. Both are soluble in organic solvents and are strong Lewis acids.
2	Beryllium hydroxide dissolves in excess of alkali and gives beryllate ion and $[Be(OH)_4]^{2-}$ and hydrogen as aluminium hydroxide which gives aluminate ion, $[Al(OH)_4]^-$.
3	Beryllium and aluminium ions have strong tendency to form complexes, BeF_4^{2-}, AlF_6^{3-}.
4	Both beryllium and aluminium hydroxides are amphoteric in nature.
5	Carbides of beryllium (Be_2C) like aluminium carbide (Al_4C_3) give methane on hydrolysis.
6	Both beryllium and aluminium are rendered passive by nitric acid.

5.5.3 Chemical properties of alkaline earth metals

The alkaline earth metals are less reactive than the alkali metals. The reactivity of these elements increases on going down the group.

Reactivity towards the halogens:

All the alkaline earth metals combine with halogen at elevated temperatures to form their halides.

$M + X_2 \rightarrow MX_2$

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